FMI	ADMINISTRATIVE INFORMATION	FMI No: 25391

FMI Title: VCT and Helios Collimator Inspection Procedure

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☑ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☑ **Staged FMI** — **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.

☐ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.

☐ **Parts Ordering**: GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator

☑ **No Parts** are required to complete this FMI.

☐ **Batch with** ______________________________

To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).

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☐ **Non-Staged** — **No Customer List is provided**. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems **Insite Center** **800-321-7937** **Milwaukee, WI 53201, USA**	**GEMS - E / ESC** **BP34** **F 78533 BUC CEDEX, FRANCE** **Tel: (33 1) 39 20 00 07 or ESC** **Fax: (33 1) 30 70 99 70**	**Asia Support Center** **Phone: 81-426-56-0033** **Fax: 81-426-48-2905**

Special Instructions:

If during the couse of performing this FMI a defectivie collimator is discovered, follow these step.

1. Inform the customer the site is down, until a new collimator is installed.
2. Submit a PSR in iTrak to report site.
3. Hold onto defectivie collimator until give instructions as to where to return it for evaluation.
4. Order replacement collimator per instuction include in FMI Inspection Procedure.

Unit Tracked: **Refer to Effectivity list on page 2.**

FMI Instructions Part No.: **5193689-100, Rev 3**

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25391 **FMI Title:** VCT and Helios Collimator Inspection Procedure

Customer Full Name:

Number and Street Address:

City and Country:

Country Code **Site Number** **System I.D.**

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified

Customer Refusal or Equipment Altered - Modification Pending.

Installer Comments

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Revision History

Revision	Date	Reason for Change
1	December 12, 2006	Initial release
2	January 30, 2007	Updated with Pilot site feedback
3	February 23, 2007	Updated Purpose section

Effectivity

The effectivity for this FMI is a select list of installed base system, tracked by either collimator or gantry serial number.

This FMI applies to all CT and PET systems that utilize the VCT (LightSpeed 7X) or Helios (LightSpeed 1X-5X) collimators that have ordered replacement FRUs (Spare Parts) between Jan. 1st, 2006 to Sept. 1st, 2006.

FRUs include complete collimator assemblies and filter sub-assemblies with the following model numbers:

Collimators	Description
5130001 (2383390)	VCT Failsafe Collimator
5140001 (2214768)	Helios Collimator
5140001-2 (2214768-2)	Helios 2 Collimator
5140001-3 (2214768-3)	Helios 3 Collimator
5140001-4 (2214768-4)	Helios 4 Collimator

Collimator Filter Assemblies (sub-FRUs)	Description
5148339	VCT Filter Assembly
2126674	Helios & -2 Filter Assembly
2291958	Helios –3 & -4 Filter Assembly

Purpose

GE Healthcare is in the process of releasing a field inspection FMI for VCT and Helios collimators used on GE CT and PET/CT scanners. This inspection is being conducted as a result of an investigation into two isolated reports out of thousands of units, from GE manufacturing, regarding collimator internal shielding.

GE Healthcare inspection and test methods for collimators include signed-off collimator manufacturer control steps, Spherical Radiation Monitoring (SRM) sample testing of collimators, and system bay staging radiation leakage testing. These redundant processes effectively identified the issue prior to product release from GE CT manufacturing.

GE Healthcare analysis of these events determined that there is not a systemic issue with collimator design or manufacture.

GE Healthcare has not received any field reports related to anomalies in collimator internal shielding from customers, service engineers, or repair centers. Collimators used for service spares are subject to the same manufacturer control sign-off as well as SRM sample testing. The vast majority of collimators are processed through manufacturing with its additional testing. GE Healthcare does not believe any units in the field are affected.

GE is proactively performing this inspection FMI on a sample of installed service spare collimators as an additional step to reconfirm and document that no shielding anomalies are in this population.

Prerequisite FMIs

FMI No.	Title
25360	Helios 221478-3 & 221478-4 Collimator X-ray Scatter Filter Bar
25367	Helios 2214768 & 2214768 - 2 Collimator X-ray Scatter Filler Bar

Related FMIs

See *Prerequisite FMIs* section above.

Furnished Materials

None.

Tool Requirements

Quantity	Description
1	3 mm Hex key socket
1	Phillips #2 screwdriver

Personnel Requirements

Resource	1 FE
Labor	2 hrs
Travel	1 hr

Before You Begin

Before beginning this procedure, please read the gantry safety information.

Safety

WARNING

POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

INSPECTION PROCESS

This FMI is for inspection of CT and PET VCT and Helios Collimators with a set effectivity. Follow all applicable instruction and return the system to normal operation.

Upon discovery of missing lead or tungsten shields in the following instructions, follow the steps below.

1	Contact: Michael L. Bauer MI&CT Service Engineering 262-312-7004 michael.l.bauer@med.ge.com
2	Submit a PSR in iTrak to report site.
3	Inform customer that the system is down until a replacement collimator is installed.
4	Hold defective collimator until given instructions as to where to send the collimator for evaluation.
5	Order replacement collimator: For VCT Collimators - Order FMI 25374 For Helios Collimators – Order through FE Tools (Spares Stock)

Procedures

It was in a VCT collimator that the missing pieces of lead and tungsten were first discovered.

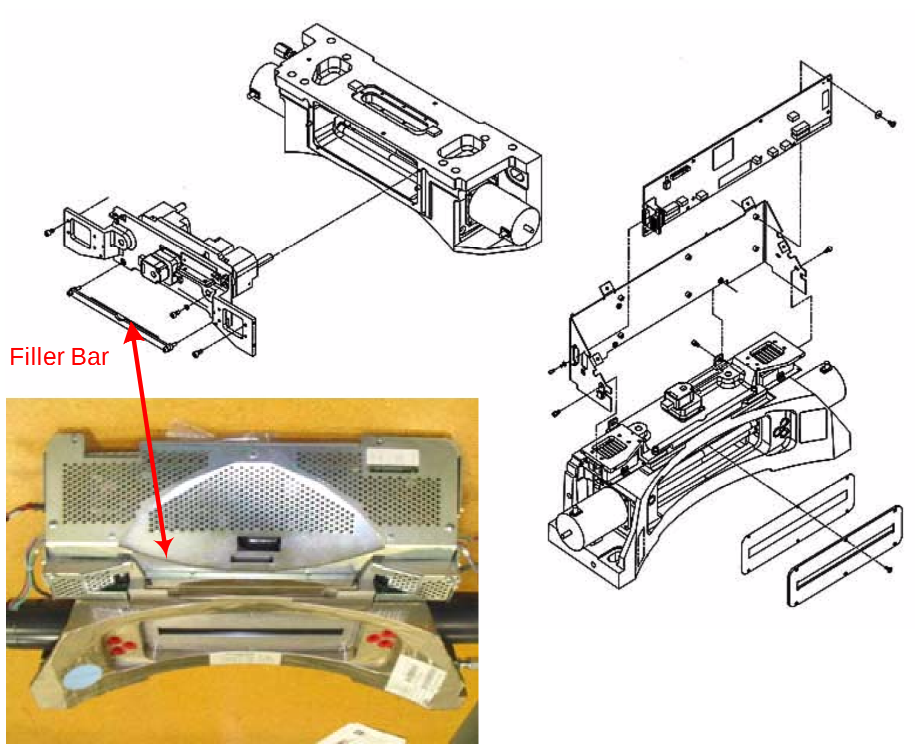
The VCT collimator is similar to the Helios collimator, which is made by same supplier. While there have been no similar problems reported with the Helios collimator, actions are being taken for the Helios collimators to be inspected as well to avoid any potential issues.

Due to certain differences in configuration, separate inspection procedures follow for the Helios and VCT collimators. If the system of interest is a VCT, proceed to the VCT section of this instruction on Page 10. Otherwise perform the following steps.

I. Helios Collimator (LightSpeed 1X-5X) Inspection Procedure

This procedure describes the process of inspecting the Helios collimator assembly.

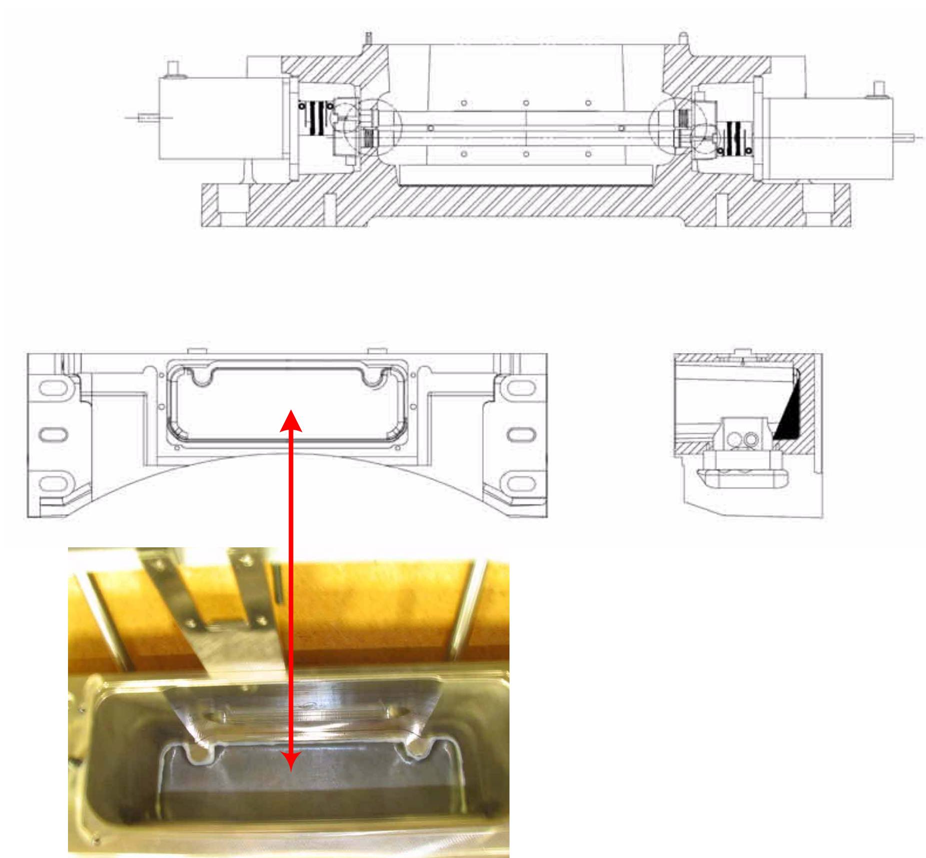
1. Shutdown system console / software.
2. Turn off system power and follow appropriate LOTO procedures.
3. Remove the gantry side, top, and front cover.
4. Position gantry so that the collimator is in the 6 o'clock position.
5. Engage gantry Rotation Lock mechanism.
6. Position a drop cloth below the collimator to collect any hardware that might be dropped.
7. Inspect for the presence of the Tungsten Filler Bar. (If found missing, return to prerequisite section and install FMI appropriate for this system.)



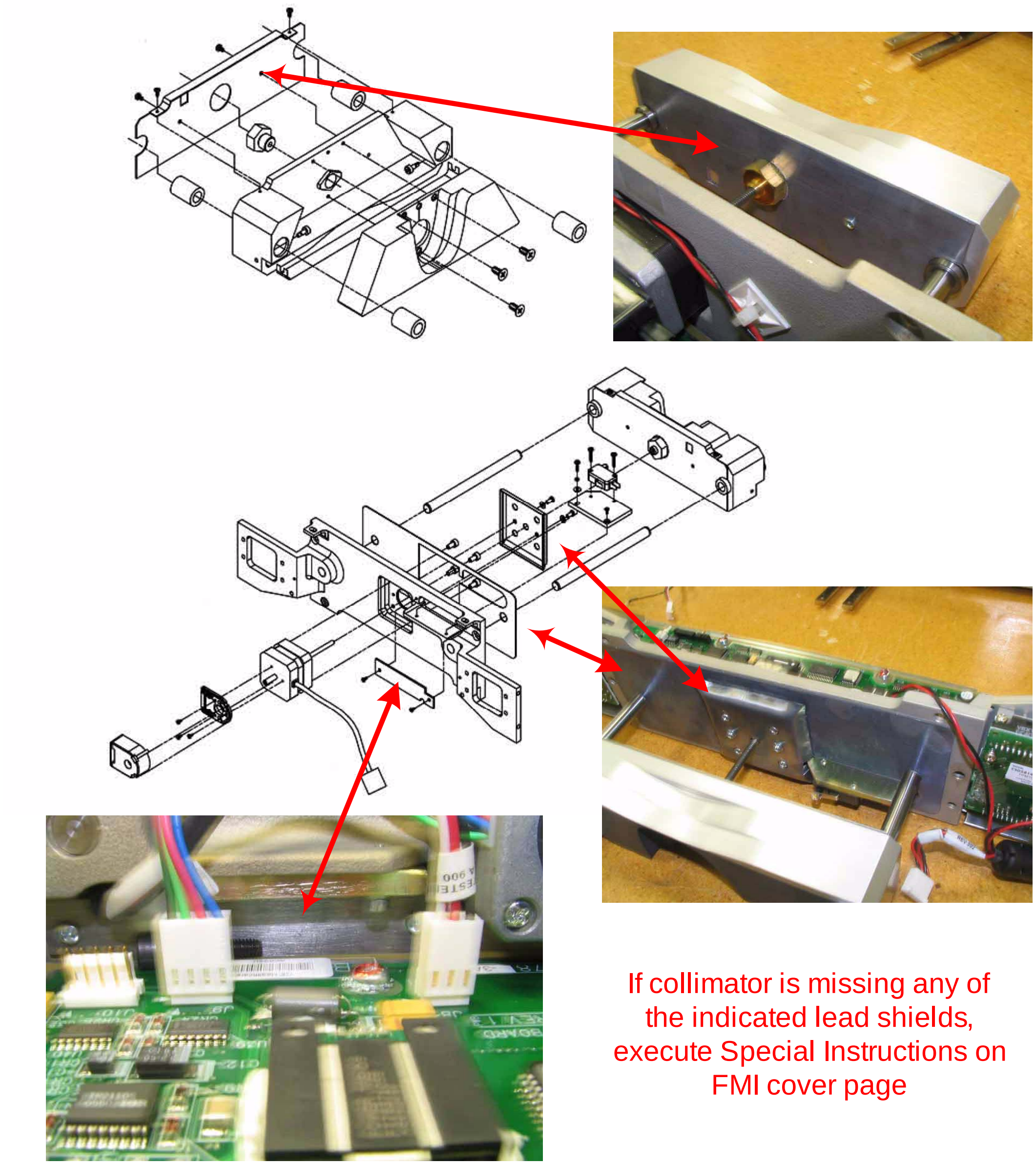
8. Follow the detailed procedure in the service manual for removing the Collimator Control Board (CCB) chassis and Collimator Filter assembly. The basic steps are these:
    a. Remove Filler Bar from collimator frame to allow removal of CCB chassis cover
    b. Remove CCB cover by removing the five pan head M4 mounting screws and set aside.
    c. Remove the CAM A and B encoder cables (J1, J2& J3) including grounding ring terminals and washers from CCB and frame mount. Return the screws and washers to their original location after removing the ring terminals.
    d. Slide the grommet on the CAM A and B encoder cables out of the frame mount and cut tie wraps to fully disconnect the encoder cables from the frame.
    e. Remove the cam motor drive covers by removing the two pan head M4 mounting screws and set aside.
    f. Disconnect the Cam A and B motor drive cables from motor drivers.
    g. Remove the two hex socket M4 frame mounting screws and carefully slide the CCB chassis and Collimator Filter assembly from the collimator base.

NOTE: *Do not damage the fragile aluminum, graphite, and copper filter assembly that extends into the collimator base during the filter assembly removal or installation.*

9. With the filter assembly removed from the collimator base, check for the presence of the lead shield in the back of the cavity in the collimator base.



10. Inspect the filter assembly for the presence of lead shields, as shown in the following illustrations.



If collimator is missing any of the indicated lead shields, execute Special Instructions on FMI cover page

11. Follow the detailed procedure in the service manual for installing the Collimator Control Board (CCB) chassis and Collimator Filter assembly. The basic steps are these:
    a. Reinsert the CCB chassis and Collimator Filter assembly in the collimator base assembly. On the filter assembly, insert the two hex socket M4 frame mounting screws and torque to 3 ± 0.3 N-m.

       NOTE: Conversion Factor: 1 N-m = 1.356 ft-lb
    b. Connect the Cam A and B motor drive cables to motor drivers.
    c. Replace the motor driver covers on the motor mount assembly; insert the two pan head M4 mounting screws and torque to 3 ± 0.3 N-m. Repeat for other motor drive cover.
    d. Connecting the Cam A and B encoder cables (J1, J2 & J3) to the CCB.
    e. Remove the screw and washer for the grounding ring terminals from the frame mount and use them secure the CAM A and B encoder grounding ring terminals and tighten.
    f. Replace the top cover of the CCB chassis using the five pan head mounting screws with spring washers and torque to 3 ± 0.3 N-m.
    g. Install the filler bar using the two M4 frame mounting screws and torque to 3 ± 0.3 N-m

12. Remove the drop cloth and follow the procedures for closing the gantry.

13. Proceed to Finalization section of this instruction on Page 12.

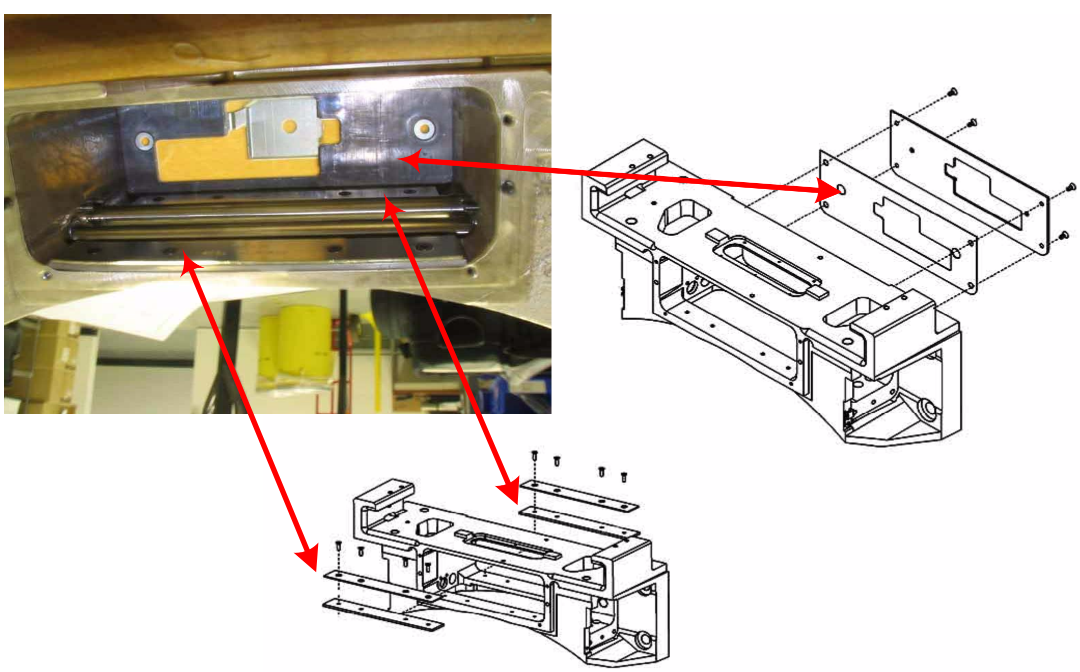
II. VCT (LightSpeed 7X) Collimator Inspection Procedure

This procedure describes the process of inspecting the VCT collimator assembly.

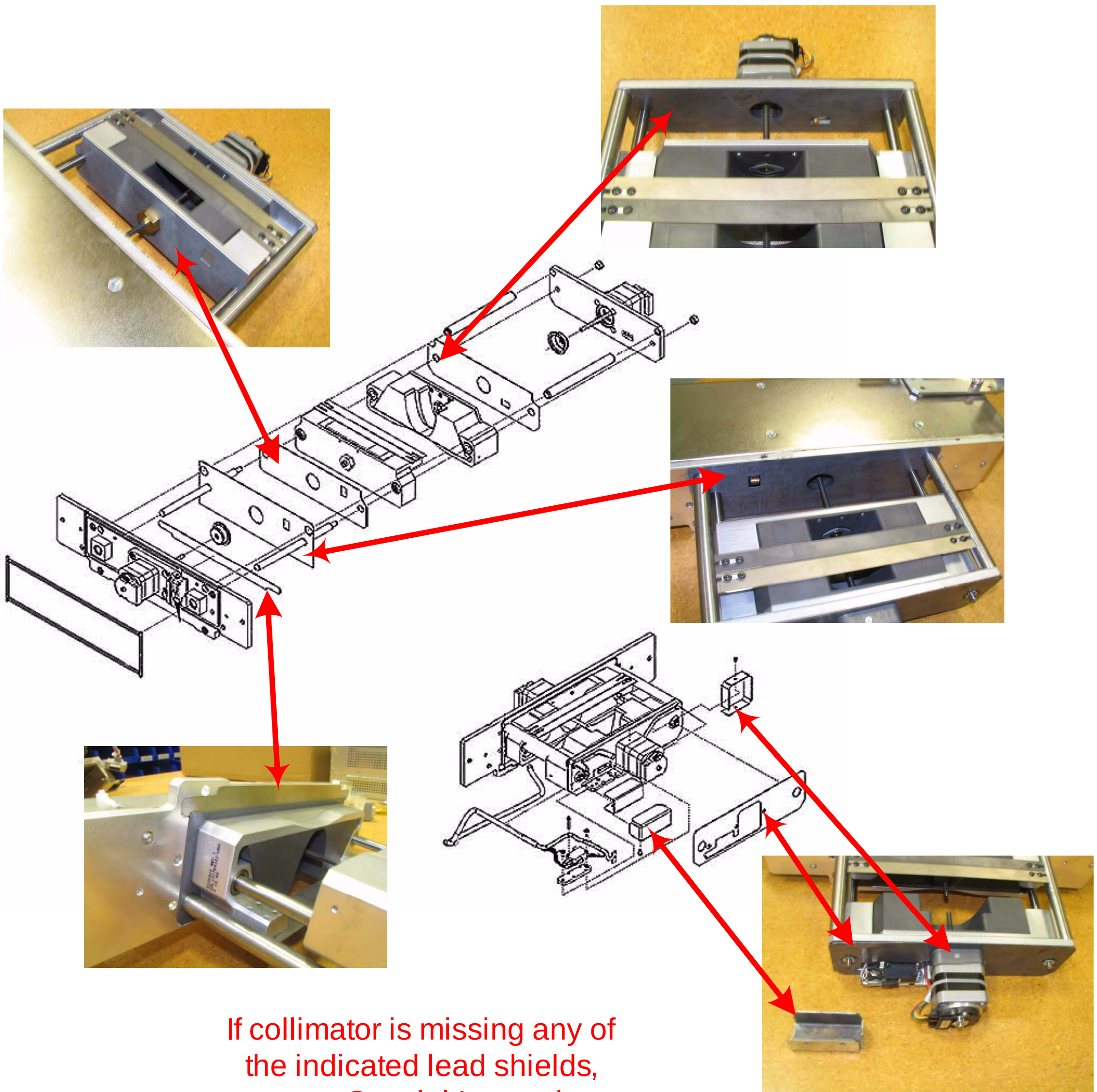
1. Shutdown system console / software.
2. Turn off system power and follow appropriate LOTO procedures.
3. Remove the gantry side, top, and front cover.
4. Position gantry so that the collimator is in the 6 o'clock position.
5. Engage gantry Rotation Lock mechanism.
6. Position a drop cloth below the collimator to collect any hardware that might be dropped.
7. Follow the detailed procedure in service manual for removing the Collimator Control Board (CCB) chassis and Collimator Filter assembly. The basic steps are these:
    a. Remove the CCB cover by removing the six pan head M4 mounting screws and set aside.
    b. Remove the CAM A and B encoder and motor drive cables from CCB chassis.
    c. Remove the four hex socket M4 frame mounting screws and carefully slide the CCB chassis and Collimator Filter assembly from the collimator base.

NOTE: *Do not damage the fragile aluminum, graphite and copper filter assembly that extends into the collimator base during the filter assembly removal or installation.*

8. With the filter assembly removed from the collimator base, check for the presence of the lead shields indicated in the following illustrations.



9. Inspect the filter assembly for the presence of lead and tungsten shields, as shown in the following illustrations.



10. Follow the detailed procedure in service manual for installing the Collimator Control Board (CCB) chassis and Collimator Filter assembly. The basic steps are these:
    a. Reinsert the CCB chassis and Collimator Filter assembly in the collimator base assembly. On the filter assembly, insert the four hex socket M4 frame mounting screws and torque to 3 ± 0.3 N-m.

       NOTE: Conversion Factor: 1 N-m = 1.356 ft-lb
    b. Connect the CAM A and B encoder and motor drive cables to CCB chassis.
    c. Replace the top cover of the CCB chassis using the six pan head mounting screws with spring washers and torque to 3 ± 0.3 N-m.

11. Remove the drop cloth and follow the procedures for closing the gantry.

12. Proceed to Finalization section of this instruction

Finalization

1. Inspect gantry for loose hardware or tools left behind.
2. Disengage Rotation Lock mechanism.
3. Replace gantry covers.
4. Remove LOTO per detailed instructions in the Service Manual and power up system.
5. With covers in place, rotate the gantry to ensure that there is no mechanical interference between the collimator and gantry cover components. Adjust covers, if required (no cover adjustment expected).
6. Perform test scan to confirm system performance before returning system for patient scanning.
7. Remove document from customer site and close FMI RFS per regional procedures.